

CentraleSupélec - ESSEC

Data Augmented Proposal Challenge

2026/02/09

Paris Office

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CentraleSupélec



The team with you for this assignment



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*Senior
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*Consultant
Consultant*



**Pietro
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*Partner
consultant*



AGENDA

1 eleven at a glance

2 A few use cases' stories

3 How to build a business proposal?

4 Case #1: The Endless Line

5 Case #2: The Next Purchase

6 Organization

ELEVEN KEY FIGURES

2008

FOUNDED IN PARIS

1 

OFFICE IN PARIS

1

GROUP

9

PARTNERS

 **+75**

CONSULTANTS

50/50

BUSINESS AND
ENGINEERS PROFILES

 **+30%**

GROWTH OVER THE LAST
24 MONTHS

+50%

ACCOMPANIED CAC40
COMPANIES

>100

ACCOMPAGNIÉ SMES



ELEVEN IS EUROPE'S FIRST EVER SPECIALIST STRATEGY CONSULTING FIRM FOUNDED TO ACCOMPANY CLIENTS' TRANSFORMATION THROUGH THE AI, DIGITAL AND CSR TRANSITION



DIGITAL STRATEGY & ACCELERATION

- Define a **strategic vision** & associated **actionable roadmap** leveraging digital & data capabilities
- Deploy **high value projects**



DATA & AI INITIATIVES IMPLEMENTATION

- **Build** and **model** problem-solving algorithms to meet **real-world needs**
- **Accelerate** the delivery of a strategic digital roadmap
- “**Excubate**” the launch of disruptive digital and data offers & services



SOCIAL & ENVIRONMENTAL RESPONSIBILITY

- Track, prove and report on **CSR impacts** within clients' initiatives, leveraging digital and AI
- Define a strategic vision for companies in **eco-responsible industries**
- Develop AI use cases to reduce **carbon footprint**



DUE DILIGENCES

- Conduct **strategic due diligences** of target/portfolio company with a digital and data angle
- Assess the **attractiveness** of a digital-enabled target. Define how to drive **digital & AI enabled value** from an asset.
- Write an **equity story** to tell. How best to position an asset **for exit**?

eleven serves top large-cap and leading mid-cap players across several key industries, as well as leading international and mid-cap Private Equity funds

Corporates

Large corporates



High-growth mid-cap



Private Equity

Generalists



Tech specialists



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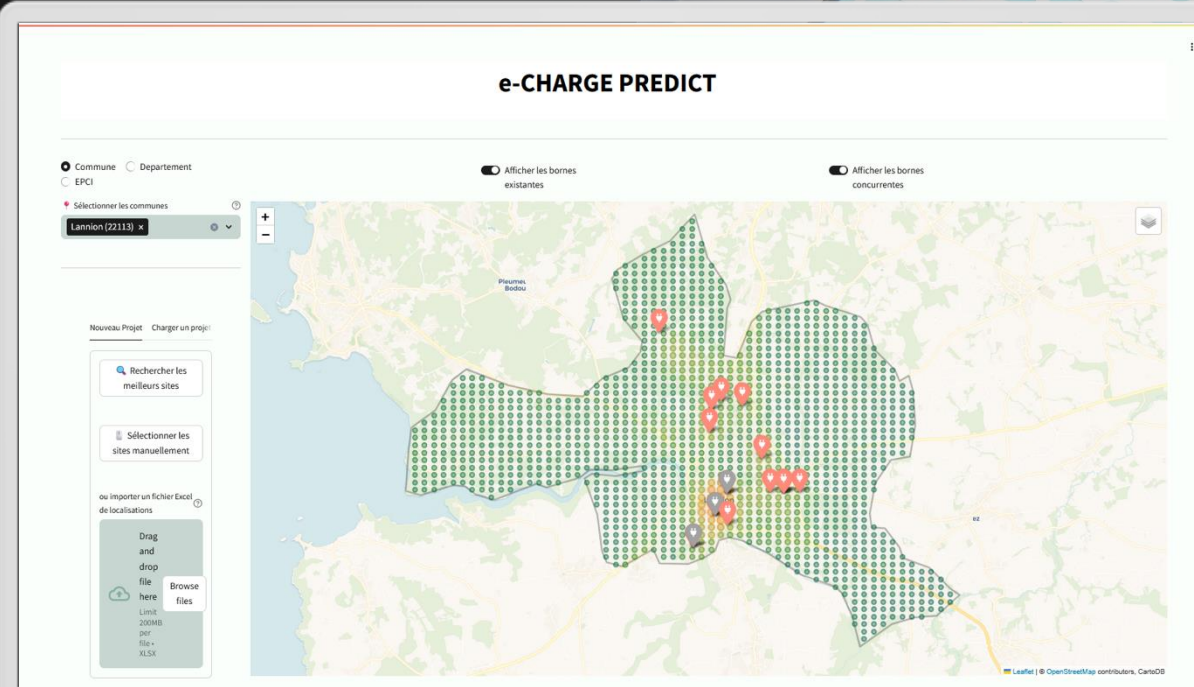
5 Case #2: The Next Purchase

6 Organization

eleven supported an EV charging operator in developing an AI-powered solution to optimize strategic deployment decisions to enhance investment profitability

EV & ML-based decision making

1 Predict EV demand to optimize charging stations location



Estimate overall **charging need** across a given area



Explore key **insights** and hypotheses



Identify **strategic** deployment **zones**



Optimize location and capacity planning

eleven optimised the management and the energy consumption of wastewater treatment plants for a major player of the French sector

Wastewater treatment and optimization

2 Optimise the management of a treatment plan with ML



Reduced energy bill by 20%, by optimizing aeration duration and timing



Increased reliability of water treatment



Created a sensor drift detection model, and virtual sensors



Built a visualization tool to monitor and pilot the plant

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Your mission: structure a data-augmented commercial exchange



Context and objectives

- Key questions: What do you understand from the project? Why is the client's need challenging?
- Present your **understanding** of the situation and client's objectives
- c. 2/3 slides



Key success factors

- What would be the reason of your success?
- Explain the reasons of success: strong **interaction** with client? Deep **expertise** in AI?
- c. 1 slide



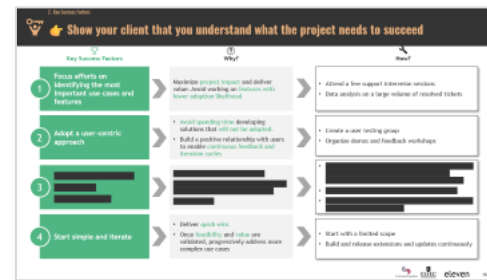
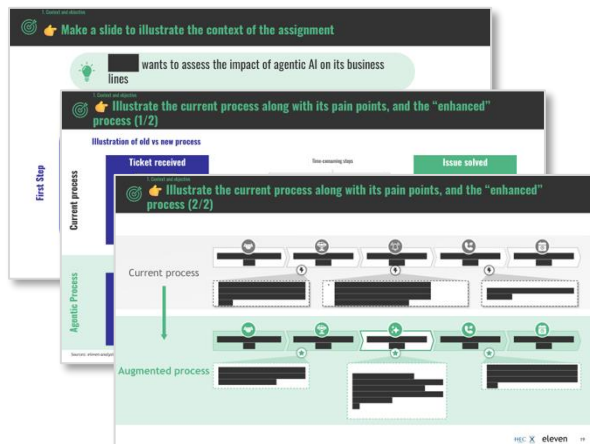
Methodology and planning

- what could you do for the potential client (implementation plan, ROI hypothesis, etc.)
- What you **could do**, **how**, what you have **already done**, any **relevant insight** to prove your expertise and **how much time** do you need **for the mission**
- Include a **demo**
- c. 6/8 slides

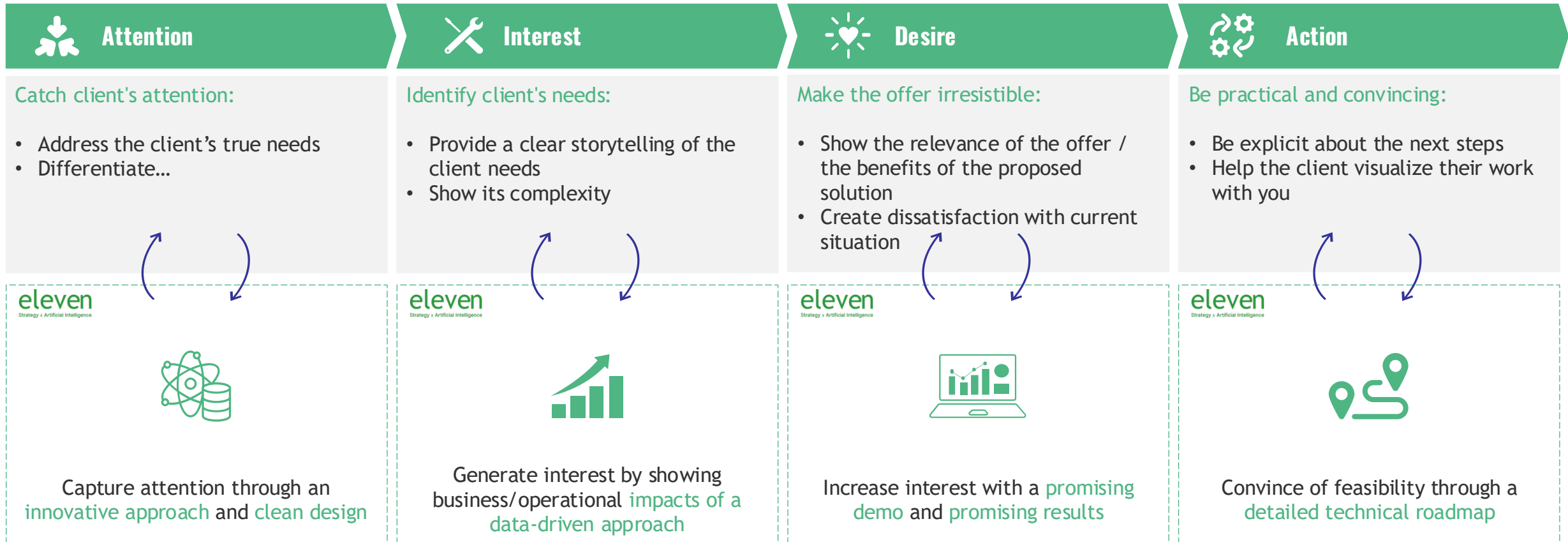


"Why you?"

- who are you and what have you done? Why are you relevant to solve the case?
- **Introduce yourself, your background** and explain **why you are relevant** for this mission
- c. 1 slide



Based on a major sales concept: **AIDA**, eleven relies on its **unique expertise** to provide **innovative** solutions to clients' problems



Key elements

Data enriches the discourse and makes it more effective at each stage of a commercial proposal



□ □ □ □ □



The IT Helpdesk:

Next steps

HR

Customer Service

Operations

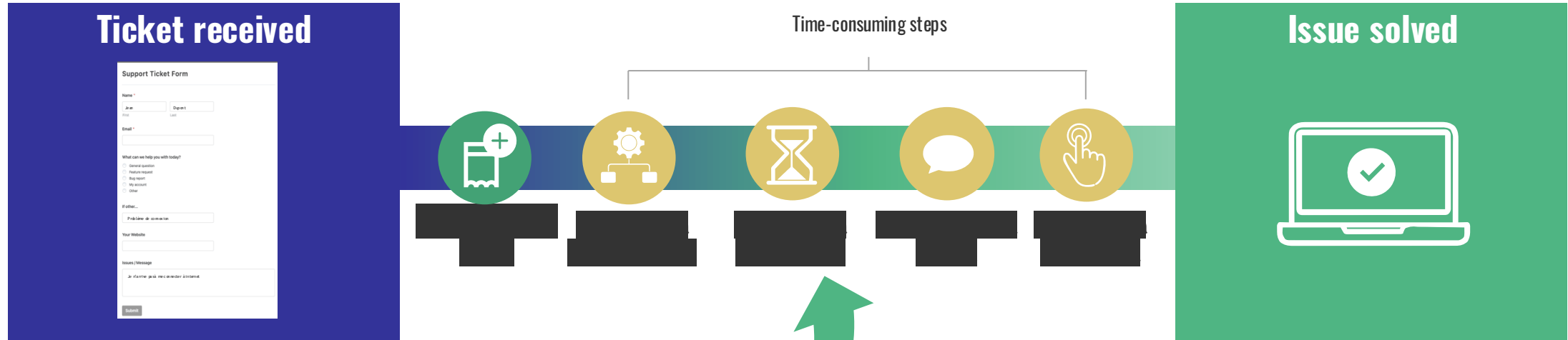
Finance



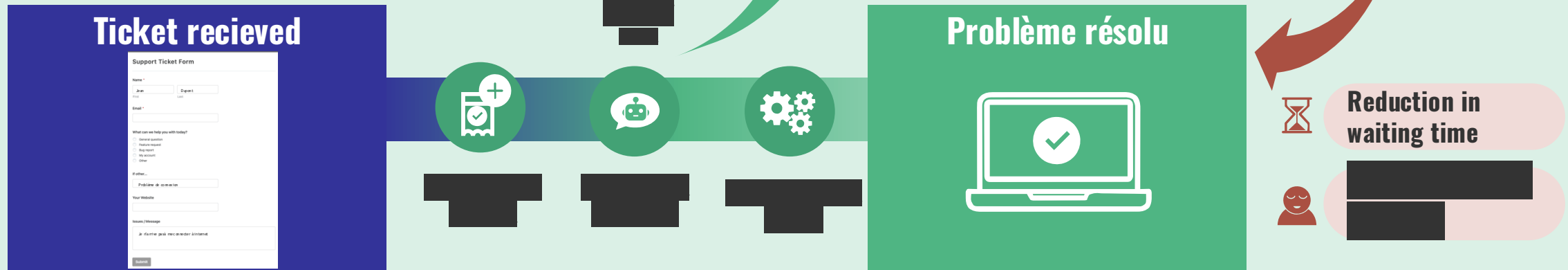
Illustrate the current process along with its pain points, and the “enhanced” process (1/2)

Illustration of old vs new process

Current process



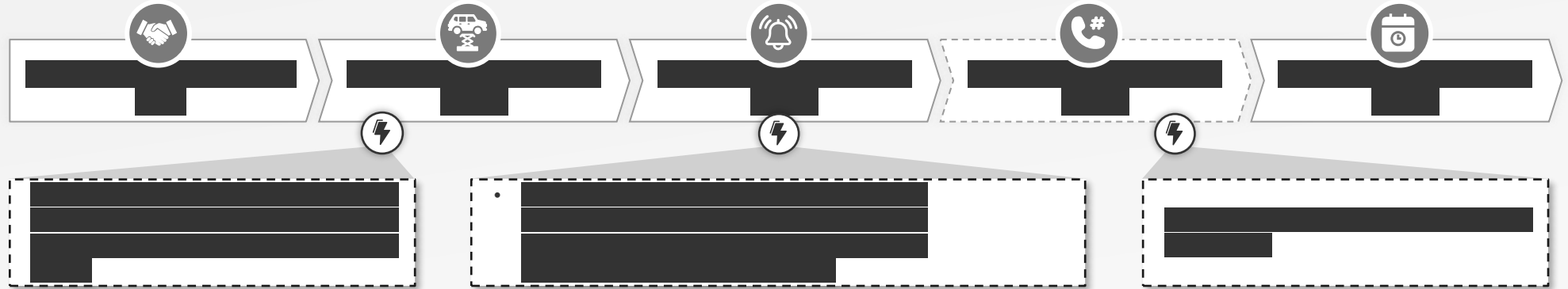
Agentic Process





Illustrate the current process along with its pain points, and the “enhanced” process (2/2)

Current process



Augmented process





Show your client that you understand what the project needs to succeed



Key Success Factors



Why?



How?

1

Focus efforts on identifying the most important use cases and features

Maximize **project impact** and deliver value. Avoid working on **features with lower adoption likelihood**

- Attend a few support intervention sessions
- Data analysis on a large volume of resolved tickets

2

Adopt a user-centric approach

- Avoid spending time developing solutions that **will not be adopted**.
- Build a positive relationship with users to enable **continuous feedback and iteration cycles**

- Create a user testing group
- Organize demos and feedback workshops

3

[Redacted]

[Redacted]

- [Redacted]
- [Redacted]
- [Redacted]

4

Start simple and iterate

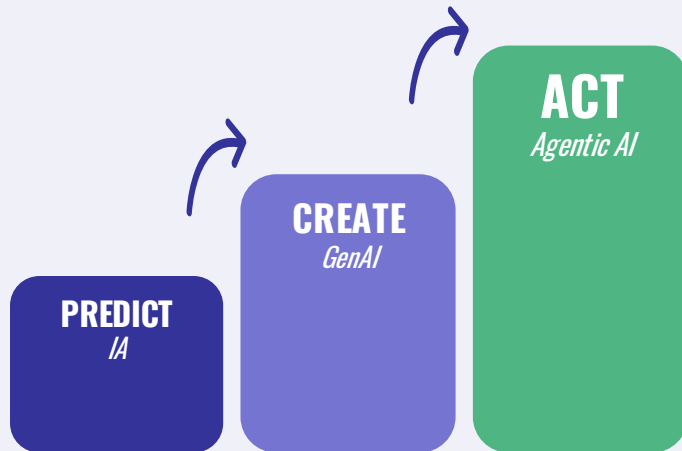
- Deliver **quick wins**
- Once **feasibility** and **value** are validated, progressively address more complex use cases

- Start with a limited scope
- Build and release extensions and updates continuously

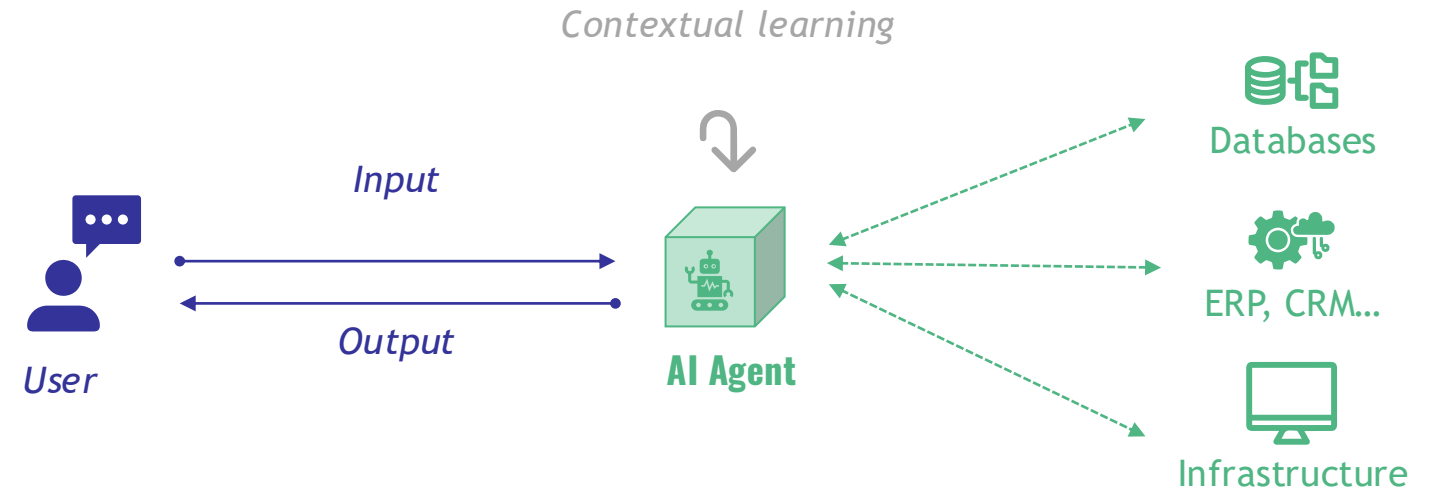


Explain to your client the key technical concepts of your methodology

Recent evolutions toward more conceptual and actionable AI...

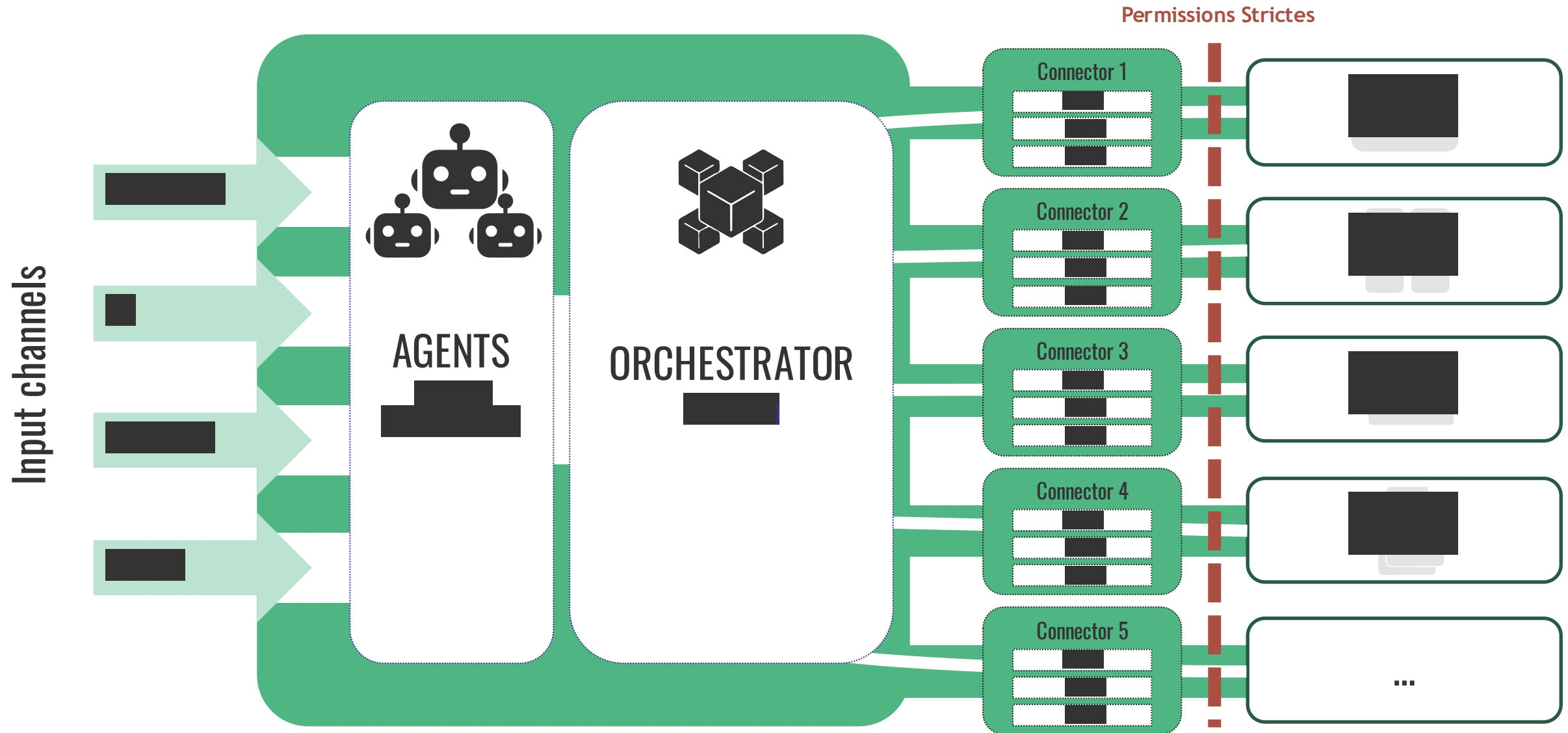


... leading to the power of agentic AI





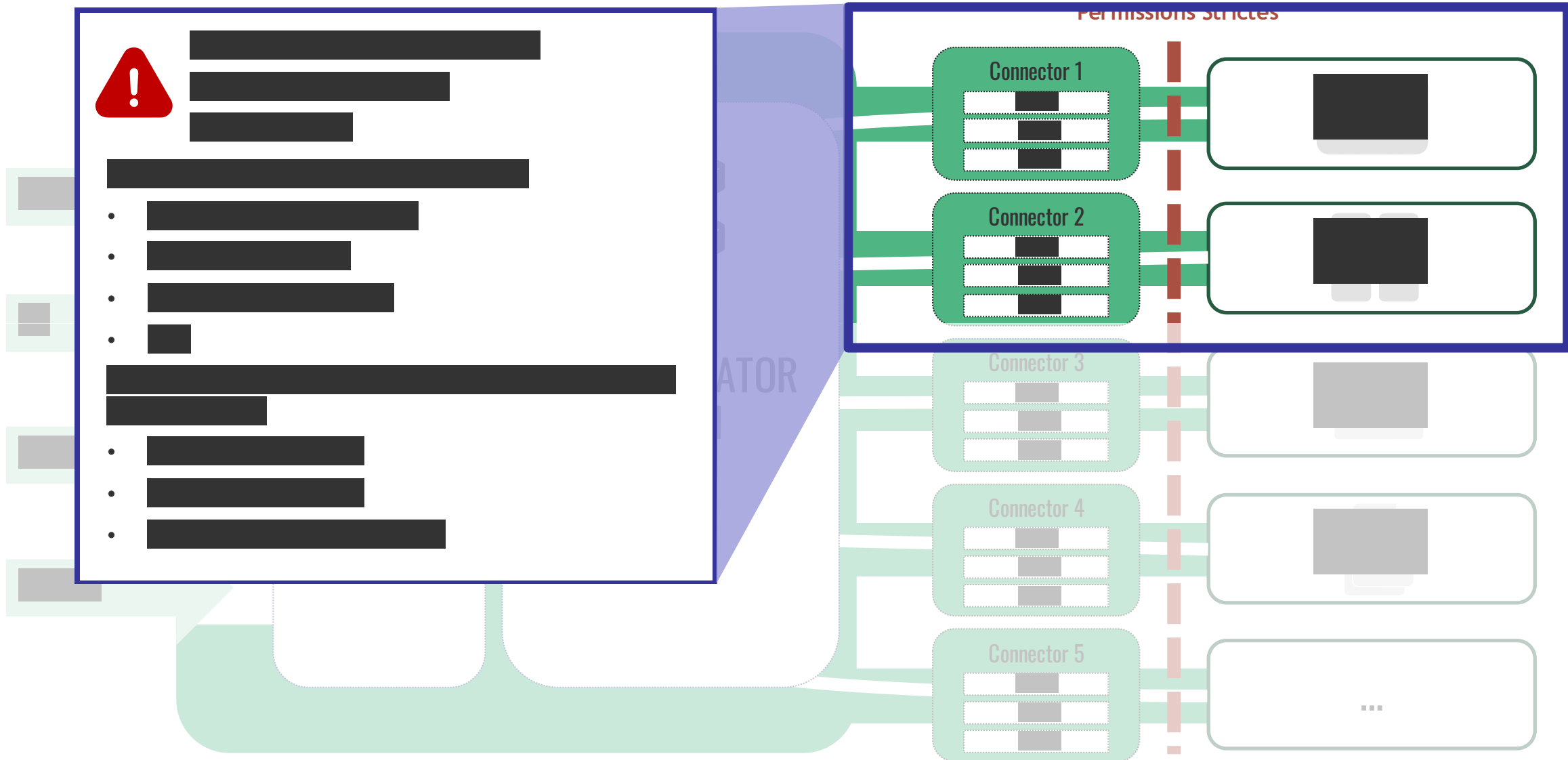
Illustrate your methodology





Flag the potential hurdles you could face during the assignment

Input channels



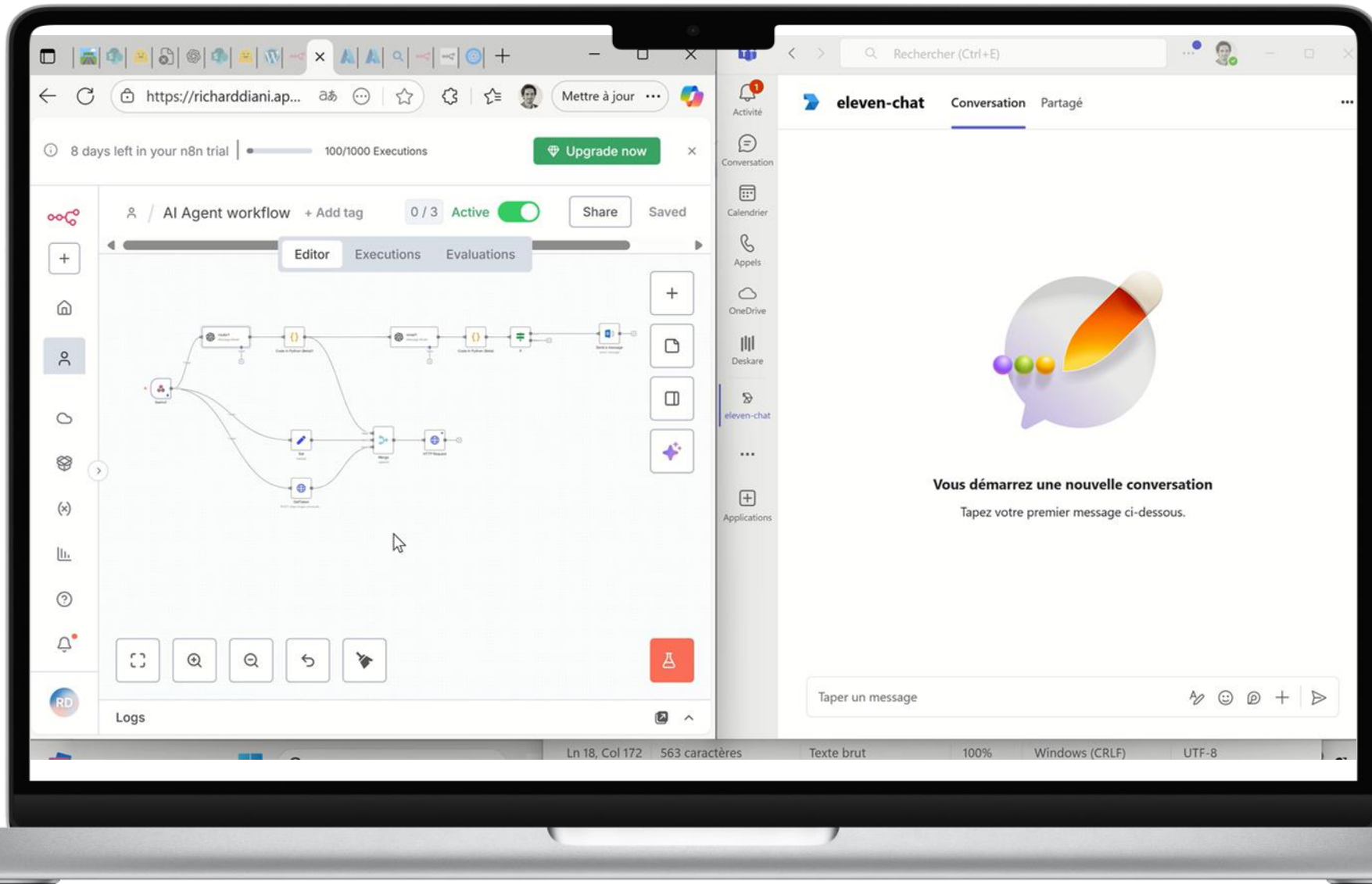


👉 Clearly define the scope of the assignment: what you will do, and what you won't

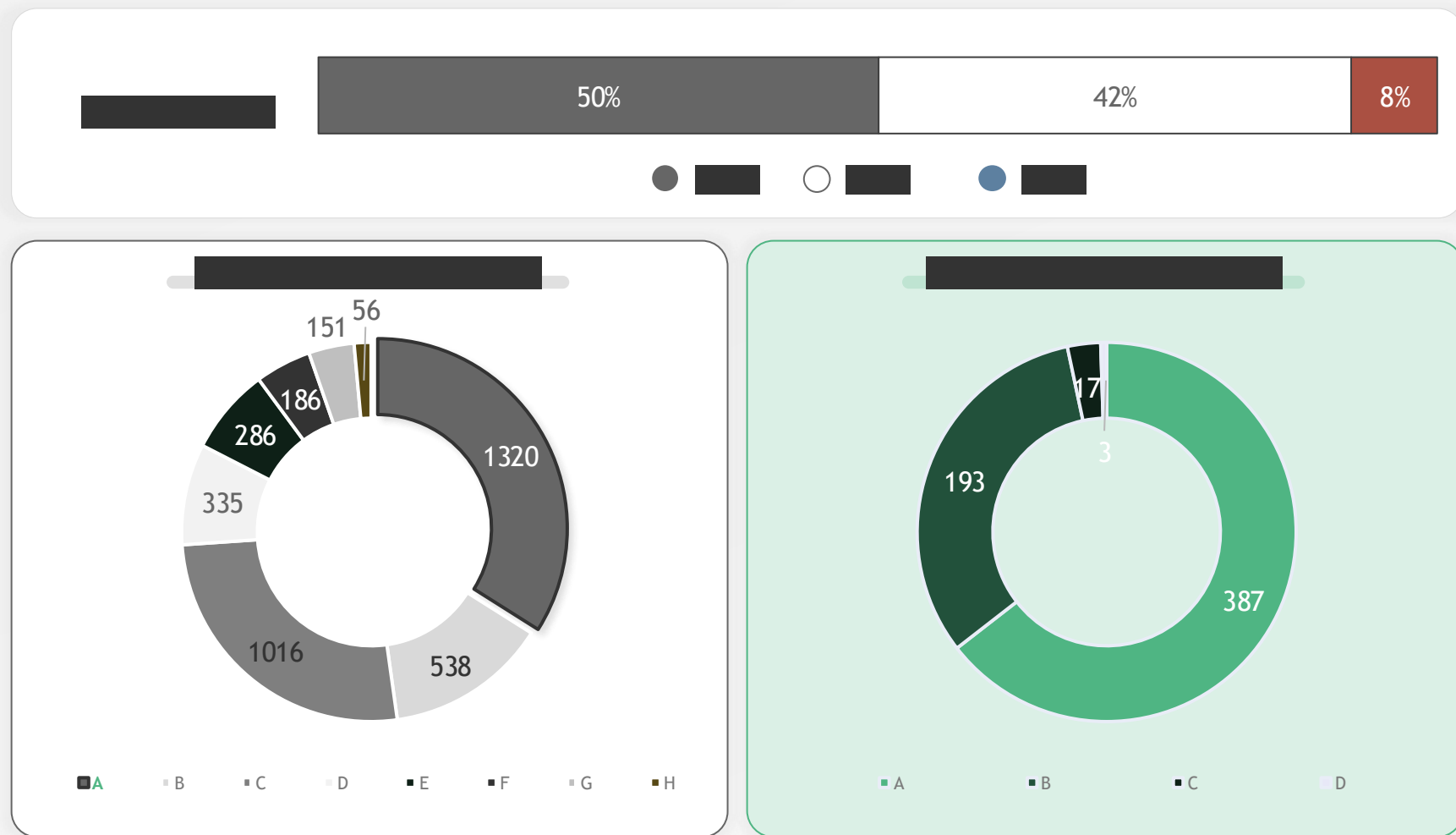




Add a preliminary demonstration



Show preliminary learnings you obtained with EDA

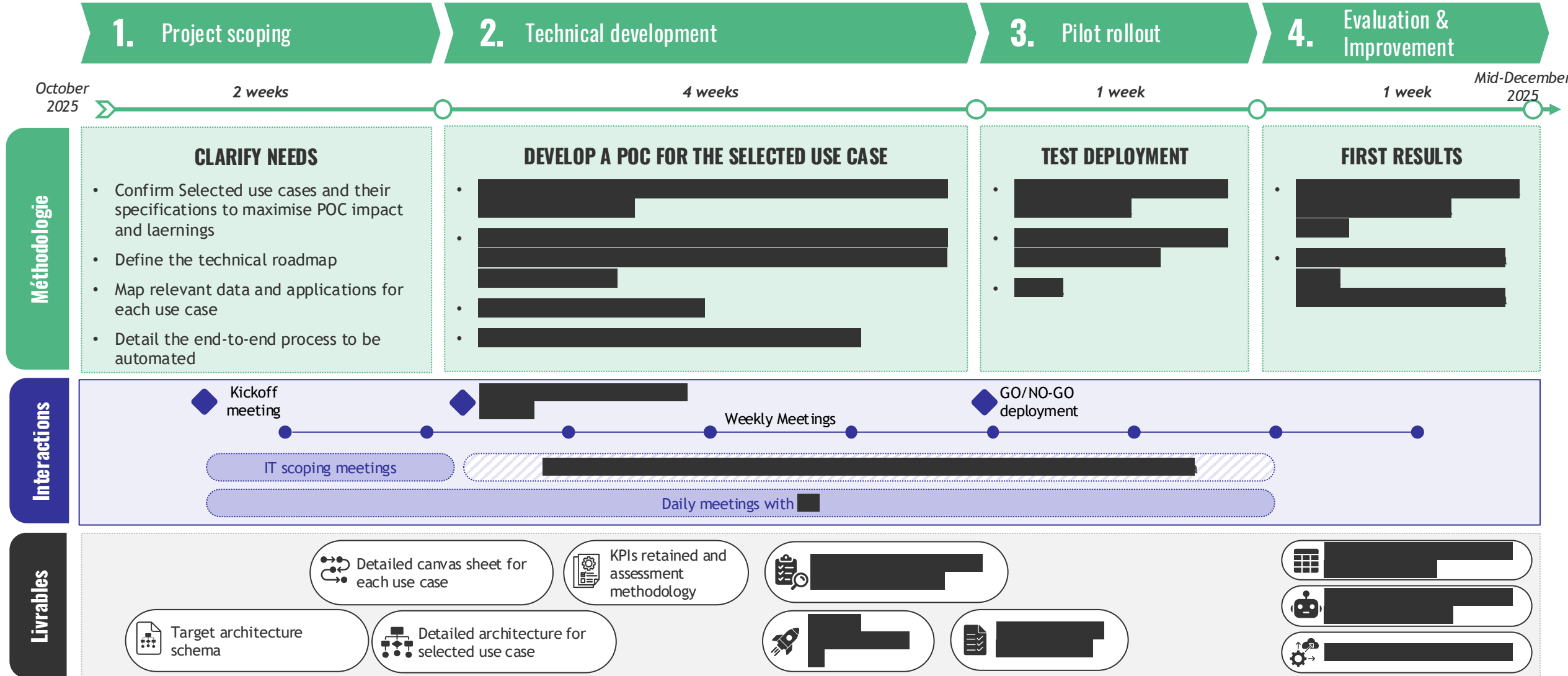


Key insights

- [Redacted]
- [Redacted]
- [Redacted]
- [Redacted]

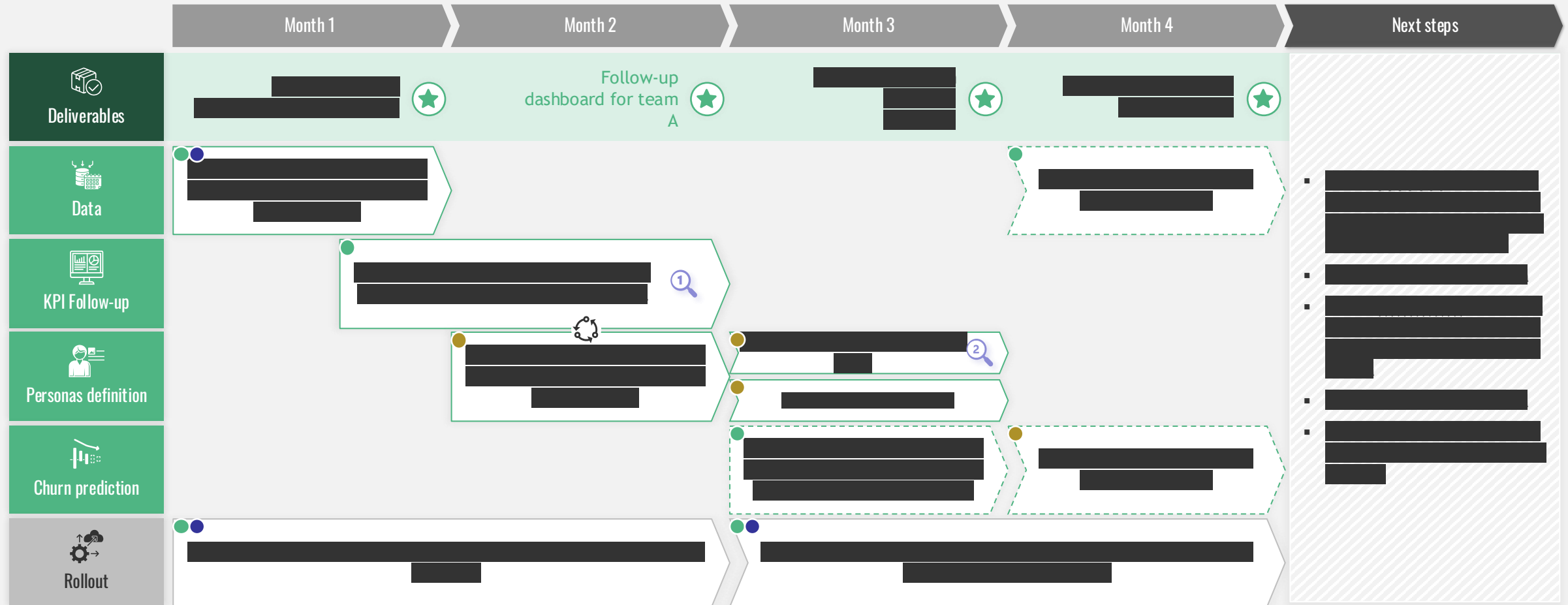


Define the key steps of the project, as well as the key interactions you will have with the client's teams and the deliverables (1/2)





Define the key steps of the project, as well as the key interactions you will have with the client's teams and the deliverables (2/2)





Explain why your team is relevant for the project, and spell out the responsibilities and expertise areas of each team member

eleven

				Client team		
	Role	Dedicated time	Responsabilities	Profile	Role	Responsabilities
Partner	 GenAI and AI expertise Strategic and technical expertise	0.5 day a week	R A C I	Project management		R A C I
Project Leader	 Strategic framing Coordination Operational follow-up of deliverables		R A C I	Innovation maagement		R A C I
Consultant(s)	 Development of AI use cases Weekly meetings		R A C I	IT management		R A C I
Assignment Total						

R

Responsible

A

Accountable

C

Consulted

I

Informed

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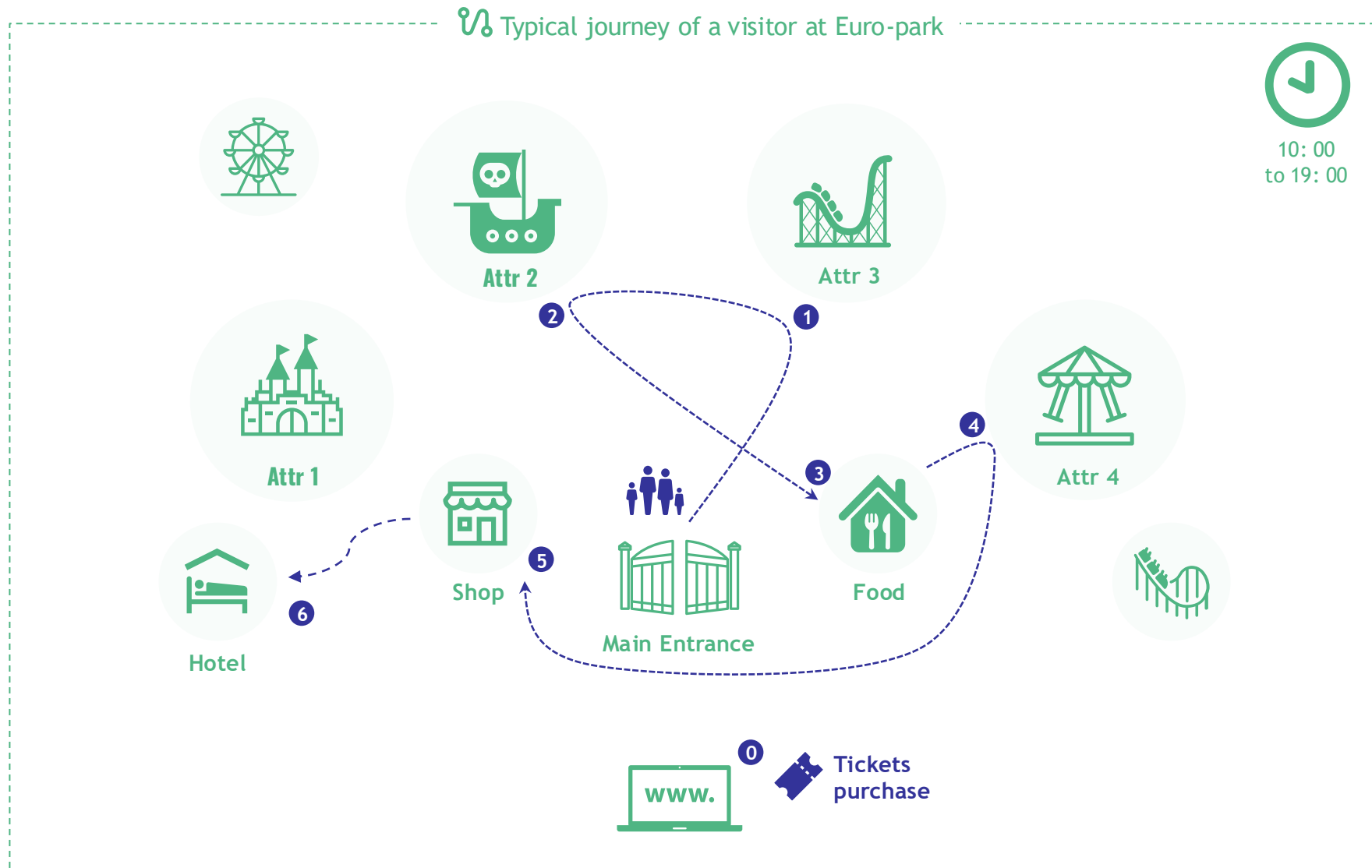
6 Organization

A low-angle shot of a roller coaster track with orange rails and green support beams, looping into a vertical curve against a clear blue sky. The track winds through the frame, creating a sense of height and motion.

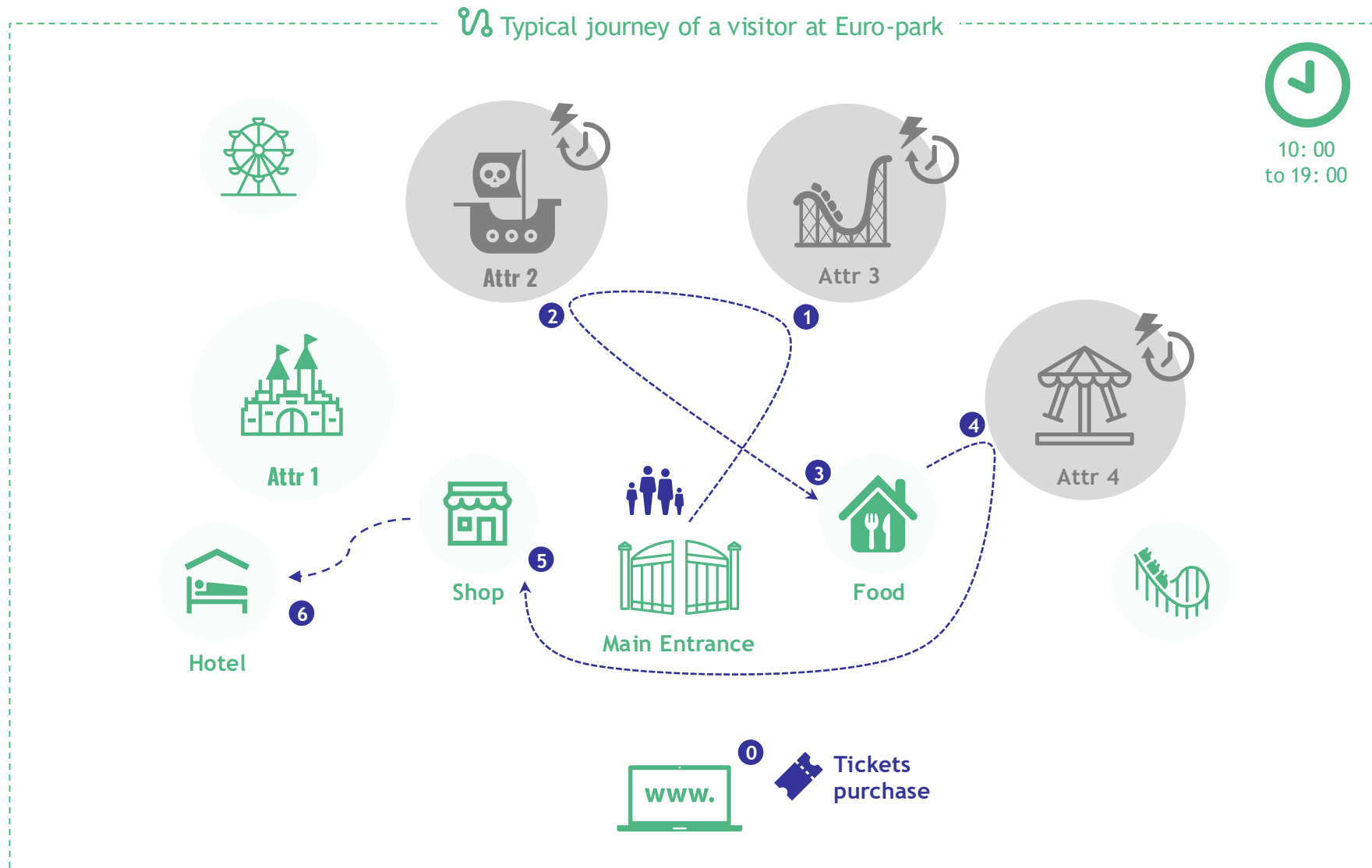
The Endless Line

Forecasting waiting times in an amusement park

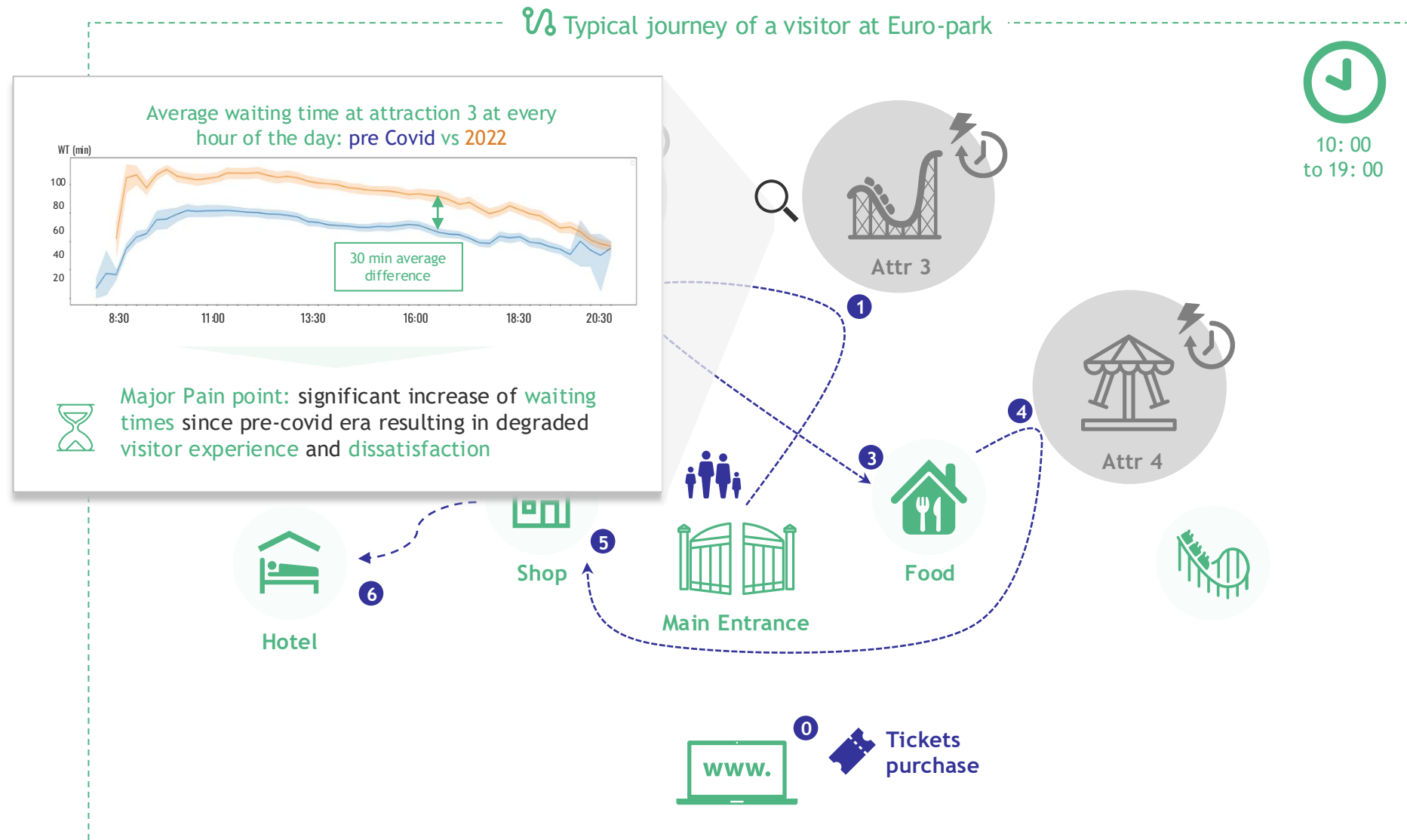
Context: Euro-park, a global theme park, is experiencing a significant increase of waiting times at attractions, which affects visitor satisfaction (1/3)



Context: Euro-park, a global theme park, is experiencing a significant increase of waiting times at attractions, which affects visitor satisfaction (2/3)



Context: Euro-park, a global theme park, is experiencing a significant increase of waiting times at attractions, which affects visitor satisfaction (3/3)



Objectives: Euro-park wants to accurately forecast waiting times for its attractions and identify use cases to leverage this information to improve the park's KPIs

1

How to accurately **forecast** attractions waiting times?

2

How to **leverage** this information to **enhance** Euro-park's KPIs?



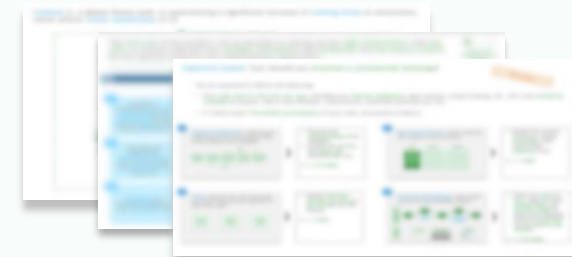
Deliverables



1. The **code** used to build the demo tool



2. A **visual dashboard** to present the results to the client



3. A "client-ready" deck of **PowerPoint slides**

Objectives: Euro-park wants to accurately forecast waiting times for its attractions and identify use cases to leverage this information to improve the park's KPIs

1

How to accurately **forecast** attractions waiting times?

2

The client is looking for **medium/long term forecasts**, assuming **normal operation** of the attractions. Therefore, make sure to remove any data related to any **unplanned event**



Deliverables



1. The **code** used to build the demo tool



2. A **visual dashboard** to present the results to the client

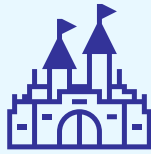


3. A "client-ready" deck of **PowerPoint slides**

Data available: you have access to past waiting times, parks' attendance, weather data, schedule of parades and opening/closing times

Euro-park operates two theme parks in the same location: we will focus on waiting times at PortAventura World

 Scope of the case study



PortAventura World



Tivoli Gardens

You have access to **five datasets** to perform your analyses and build your models



waiting_times.csv



attendance.csv



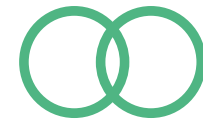
entity_schedule.csv



weather.csv



parade_night_show.csv



link_attraction_park.csv

Data available: past waiting times are provided for each attraction at a granularity of 15 minutes

1

15 minutes time slots

Waiting time

WORK_DATE	DEB_TIME	DEB_TIME_HOUR	FIN_TIME	ENTITY_DESCRIPTION_SHORT	WAIT_TIME_MAX	NB_UNITS	GUEST_CARRIED	CAPACITY	ADJUST_CAPACITY	OPEN_TIME	UP_TIME	DOWNTIME	NB_MAX_UNIT
01/01/2018	01/01/2018 21:00	21	01/01/2018 21:15	Roller Coaster	0	2.0	0.0	0.0	0.0	0	0	0	0 2.0
01/01/2018	01/01/2018 19:30	19	01/01/2018 19:45	Bumper Cars	5	18.0	148.0	254.749	254.75	15	15	15	0 18.0
01/01/2018	01/01/2018 22:30	22	01/01/2018 22:45	Rapids Ride	0	1.0	0.0	0.0	0.0	0	0	0	0 2.0
01/01/2018	01/01/2018 12:45	12	01/01/2018 13:00	Crazy Dance	5	1.0	46.0	250.001	250.0	15	15	15	0 1.0
01/01/2018	01/01/2018 17:00	17	01/01/2018 17:15	Skyway	5	15.0	92.0	211.5	198.25	15	15	15	0 16.0
01/01/2018	01/01/2018 18:15	18	01/01/2018 18:30	Free Fall	50	3.0	0.0	0.0	0.0	0	0	0	0 3.0
01/01/2018	01/01/2018 13:30	13	01/01/2018 13:45	Monorail	70	11.0	145.0	223.751	223.75	15	15	15	0 11.0
01/01/2018	01/01/2018 15:00	15	01/01/2018 15:15	Roller Coaster	20	2.0	51.0	75.0	75.0	15	15	15	0 2.0
01/01/2018	01/01/2018 18:00	18	01/01/2018 18:15	Swing Ride	50	12.0	74.0	242.25	242.25	15	15	15	0 12.0
01/01/2018	01/01/2018 17:15	17	01/01/2018 17:30	Crazy Bus	5	6.0	271.0	353.0	353.0	15	15	15	0 6.0
01/01/2018	01/01/2018 13:15	13	01/01/2018 13:30	Drop Tower	5	16.0	41.9999	140.25	140.25	15	15	15	0 16.0
01/01/2018	01/01/2018 18:45	18	01/01/2018 19:00	Spinning Coaster	45	6.0	309.0	526.25	526.25	15	15	15	0 6.0
01/01/2018	01/01/2018 21:00	21	01/01/2018 21:15	Monorail	0	11.0	0.0	0.0	0.0	0	0	0	0 11.0
01/01/2018	01/01/2018 11:30	11	01/01/2018 11:45	Scooby Doo	60	36.0	162.0	425.0	425.0	15	15	15	0 36.0
01/01/2018	01/01/2018 19:30	19	01/01/2018 19:45	Superman Ride	0	2.0	0.0	0.0	0.0	0	0	0	0 3.0
01/01/2018	01/01/2018 17:30	17	01/01/2018 17:45	Spiral Slide	0	2.0	37.0	75.0	75.0	15	15	15	0 2.0
01/01/2018	01/01/2018 22:30	22	01/01/2018 22:45	Inverted Coaster	0	1.0	0.0	0.0	0.0	0	0	0	0 3.0
01/01/2018	01/01/2018 11:00	11	01/01/2018 11:15	Spinning Coaster	25	5.0	230.0	526.25	438.5	15	15	15	0 6.0
01/01/2018	01/01/2018 21:30	21	01/01/2018 21:45	Water Ride	15	10.0	0.0	0.0	0.0	0	0	0	0 11.0
01/01/2018	01/01/2018 12:45	12	01/01/2018 13:00	Water Ride	20	10.0	133.0	247.001	224.5	15	15	15	0 11.0
01/01/2018	01/01/2018 13:15	13	01/01/2018 13:30	Power Tower	5	16.0	47.0	234.499	220.75	15	15	15	0 17.0
01/01/2018	01/01/2018 20:15	20	01/01/2018 20:30	Roller Coaster	0	2.0	0.0	0.0	0.0	0	0	0	0 2.0
01/01/2018	01/01/2018 13:15	13	01/01/2018 13:30	Free Fall	90	3.0	61.0	134.749	134.75	15	15	15	0 3.0
01/01/2018	01/01/2018 11:30	11	01/01/2018 11:45	Water Ride	15	10.0	130.0	247.001	224.5	15	15	15	0 11.0
01/01/2018	01/01/2018 11:00	11	01/01/2018 11:15	Top Spin	10	1.0	36.0	288.25	144.25	15	15	15	0 2.0
01/01/2018	01/01/2018 17:00	17	01/01/2018 17:15	Crazy Bus	5	6.0	264.0	353.0	353.0	15	15	15	0 6.0
01/01/2018	01/01/2018 12:15	12	01/01/2018 12:30	Log Flume	50	87.0	173.0	429.5	424.5	15	15	15	0 88.0
01/01/2018	01/01/2018 15:30	15	01/01/2018 15:45	Oz Theatre	10	2.0	80.0	387.5	387.5	15	15	15	0 2.0
01/01/2018	01/01/2018 18:45	18	01/01/2018 19:00	Circus Train	0	1.0	0.0	0.0	0.0	0	0	0	0 1.0
01/01/2018	01/01/2018 11:15	11	01/01/2018 11:30	Giant Wheel	55	3.0	188.0	503.75	302.25	15	15	15	0 5.0

Data available: you also have access to the park's daily overall attendance

2

USAGE_DATE ▼	FACILITY_NAME ▼	attendance ▼
01/06/2018	PortAventura World	46804
01/06/2018	Tivoli Gardens	20420
02/06/2018	PortAventura World	57940
02/06/2018	Tivoli Gardens	29110
03/06/2018	PortAventura World	44365
03/06/2018	Tivoli Gardens	23727
04/06/2018	PortAventura World	37617
04/06/2018	Tivoli Gardens	15115
05/06/2018	PortAventura World	32438
05/06/2018	Tivoli Gardens	16373
06/06/2018	PortAventura World	28399
06/06/2018	Tivoli Gardens	15831
07/06/2018	PortAventura World	39436
07/06/2018	Tivoli Gardens	16587
08/06/2018	PortAventura World	36459

Data available: The entity_schedule dataset contains all information on the opening, closing and rehabilitation works of attractions

3

REF_CLOSING_DESCRIPTION ▼	ENTITY_DESCRIPTION_SHORT ▼	ENTITY_TYPE ▼	DEB_TIME ▼	FIN_TIME ▼	UPDATE_TIME ▼	WORK_DATE ▼
	Tivoli Gardens	PARK	19/11/2018 10:00	19/11/2018 18:00	20/11/2018 08:24	19/11/2018
	Dizzy Dropper	ATTR	07/04/2022 08:30	07/04/2022 22:04	08/04/2022 08:00	07/04/2022
	Sling Shot	ATTR	28/03/2018 08:37	28/03/2018 18:12	29/03/2018 08:24	28/03/2018
	Gondola	ATTR	11/04/2019 09:55	11/04/2019 20:19	12/04/2019 08:59	11/04/2019
	Monorail	ATTR	29/06/2019 08:30	29/06/2019 20:35	30/06/2019 08:14	29/06/2019
Fermeture Réhab	Top Spin	ATTR	07/04/2019 23:59	07/04/2019 23:59	08/04/2019 07:40	07/04/2019
	PortAventura World	PARK	17/07/2018 08:30	17/07/2018 09:30	18/07/2018 07:48	17/07/2018
	Skyway	ATTR	13/12/2019 08:23	13/12/2019 18:02	14/12/2019 07:59	13/12/2019
	Sling Shot	ATTR	21/07/2018 08:33	21/07/2018 20:13	22/07/2018 07:46	21/07/2018
	Gondola	ATTR	11/11/2019 09:00	11/11/2019 18:22	12/11/2019 07:18	11/11/2019
	Tivoli Gardens	PARK	19/04/2019 08:30	19/04/2019 09:30	20/04/2019 07:43	19/04/2019
	PortAventura World	PARK	28/02/2019 08:30	28/02/2019 09:30	01/03/2019 09:23	28/02/2019
Fermeture Réhab	Kiddie Coaster	ATTR	21/02/2018 23:59	21/02/2018 23:59	22/02/2018 08:36	21/02/2018
	PortAventura World	PARK	18/08/2019 08:30	18/08/2019 09:30	19/08/2019 07:34	18/08/2019
Fermeture Opérationnelle	Oz Theatre	ATTR	04/09/2018 23:59	04/09/2018 23:59	05/09/2018 08:12	04/09/2018
	PortAventura World	PARK	05/02/2020 10:00	05/02/2020 19:00	06/02/2020 08:09	05/02/2020
	Free Fall	ATTR	04/01/2020 08:11	04/01/2020 18:15	05/01/2020 07:52	04/01/2020
	Tivoli Gardens	PARK	04/05/2018 08:30	04/05/2018 09:30	05/05/2018 11:16	04/05/2018
Fermeture Opérationnelle	Oz Theatre	ATTR	24/06/2018 23:59	24/06/2018 23:59	25/06/2018 08:03	24/06/2018
	Giga Coaster	ATTR	08/04/2019 09:25	08/04/2019 21:00	09/04/2019 07:45	08/04/2019
	Tivoli Gardens	PARK	29/01/2020 08:30	29/01/2020 09:30	03/02/2020 15:22	29/01/2020
	Inverted Coaster	ATTR	15/09/2018 09:55	15/09/2018 18:20	16/09/2018 08:38	15/09/2018
	Himalaya Ride	ATTR	05/06/2018 10:00	05/06/2018 23:00	06/06/2018 07:27	05/06/2018
	Water Ride	ATTR	02/05/2019 08:30	02/05/2019 21:14	05/06/2019 09:12	02/05/2019

Data available: The hourly weather at the park was also made available

4

dt_iso	temp	visibility	feels_like	temp_min	temp_max	wind_gust	rain_1h	rain_3h	snow_1h	snow_3h	clouds_all	weather_main	weather_description	weather_icon
1999-01-01 00:00:00 +0000 UTC	8.33		5.28	8.14	9.32						8	Clear	sky is clear	01n
1999-01-01 01:00:00 +0000 UTC	8.08		5.18	7.94	9.32						6	Clear	sky is clear	01n
1999-01-01 02:00:00 +0000 UTC	8.08		5.38	7.94	8.74						14	Clouds	few clouds	02n
1999-01-01 03:00:00 +0000 UTC	7.31		4.42	7.14	8.62						39	Clouds	scattered clouds	03n
1999-01-01 04:00:00 +0000 UTC	6.91		4.0	6.74	8.32						52	Clouds	broken clouds	04n
1999-01-01 05:00:00 +0000 UTC	5.45		2.35	4.6	7.32						52	Clouds	broken clouds	04n
1999-01-01 06:00:00 +0000 UTC	5.63		2.57	4.6	7.62						57	Clouds	broken clouds	04n
1999-01-01 07:00:00 +0000 UTC	5.64		2.38	3.6	7.32						65	Clouds	broken clouds	04n
1999-01-01 08:00:00 +0000 UTC	5.99		2.69	4.6	6.94						61	Clouds	broken clouds	04d
1999-01-01 09:00:00 +0000 UTC	6.72		3.42	5.6	7.97						83	Clouds	broken clouds	04d
1999-01-01 10:00:00 +0000 UTC	7.01		3.97	5.6	8.77						89	Clouds	overcast clouds	04d
1999-01-01 11:00:00 +0000 UTC	9.41		7.22	8.6	11.77						92	Clouds	overcast clouds	04d
1999-01-01 12:00:00 +0000 UTC	10.98		10.14	10.6	11.47						73	Clouds	broken clouds	04d
1999-01-01 13:00:00 +0000 UTC	10.8		9.92	10.6	12.32						32	Clouds	scattered clouds	03d
1999-01-01 14:00:00 +0000 UTC	12.09		11.34	11.32	12.77						52	Clouds	broken clouds	04d
1999-01-01 15:00:00 +0000 UTC	11.6		10.83	10.92	11.94						42	Clouds	scattered clouds	03d
1999-01-01 16:00:00 +0000 UTC	10.38		9.61	8.77	10.94						32	Clouds	scattered clouds	03d
1999-01-01 17:00:00 +0000 UTC	9.69		7.88	8.32	9.94						8	Clear	sky is clear	01n
1999-01-01 18:00:00 +0000 UTC	7.93		5.59	6.07	8.6						8	Clear	sky is clear	01n
1999-01-01 19:00:00 +0000 UTC	7.69		5.29	6.74	8.32						19	Clouds	few clouds	02n
1999-01-01 20:00:00 +0000 UTC	6.8		4.17	6.6	7.32						79	Clouds	broken clouds	04n
1999-01-01 21:00:00 +0000 UTC	6.56		3.86	6.14	6.72						100	Clouds	overcast clouds	04n
1999-01-01 22:00:00 +0000 UTC	5.91		2.97	5.74	6.32						98	Clouds	overcast clouds	04n
1999-01-01 23:00:00 +0000 UTC	5.91		2.8	5.74	6.32						94	Clouds	overcast clouds	04n
1999-01-02 00:00:00 +0000 UTC	5.78		2.29	5.44	6.32						100	Clouds	overcast clouds	04n
1999-01-02 01:00:00 +0000 UTC	5.91		2.2	5.74	6.32						100	Clouds	overcast clouds	04n
1999-01-02 02:00:00 +0000 UTC	6.74		3.04	5.74	7.32						100	Clouds	overcast clouds	04n
1999-01-02 03:00:00 +0000 UTC	6.64		2.74	6.64	6.92						100	Clouds	overcast clouds	04n
1999-01-02 04:00:00 +0000 UTC	6.91		2.94	6.74	7.32						100	Clouds	overcast clouds	04n

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2 A few use cases' stories

3 How to build a business proposal?

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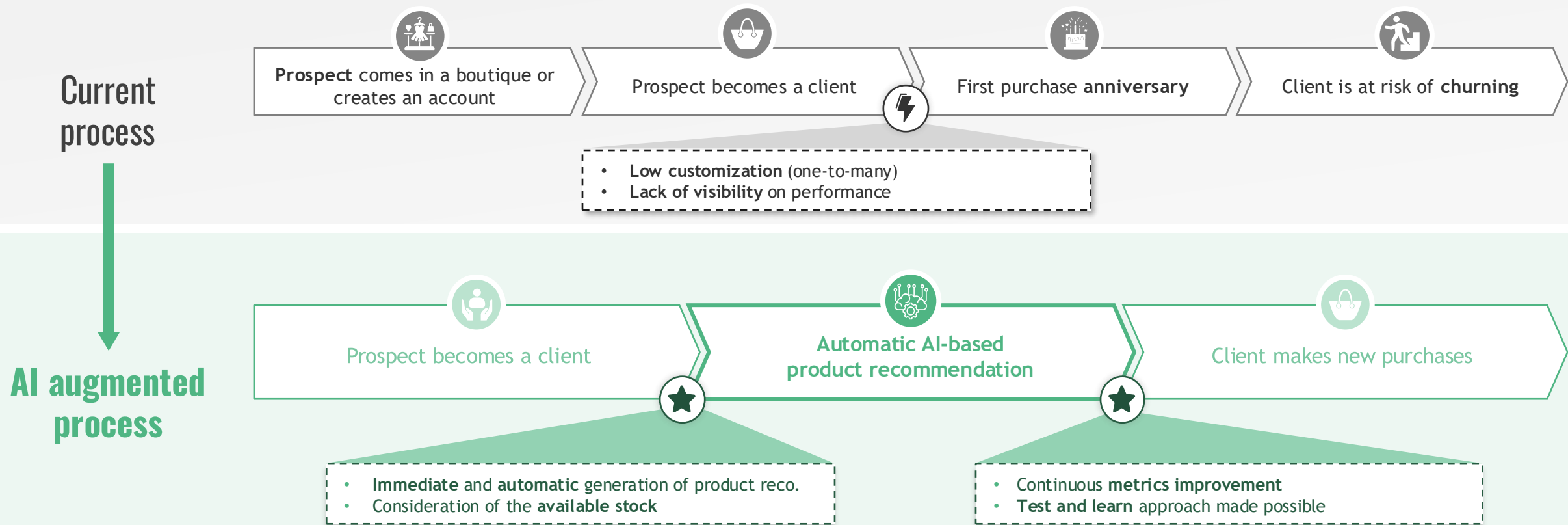
6 Organization

An aerial, high-angle view of a large, modern stadium at night. The stadium is illuminated by bright lights, and the pitch is a vibrant blue. The seating areas are filled with yellow seats, and the roof structure is visible above the stadium. The overall atmosphere is one of a major sporting event or a significant announcement.

The Next Purchase

**Recommending the right product to each client
for a marketing action**

Context: the potential client, a player of the retail industry, **wants to improve the performance of marketing actions** (product recommendation) based on artificial intelligence



Key benefits

- Push the **right product** recommendation to the **right client** thus increasing number of repeaters
- **Save time** by automating the creation of product recommendations

Objectives: The client wants to **recommend products relevant to each client** to customize its marketing actions and increase revenues

Scope of the case study

1

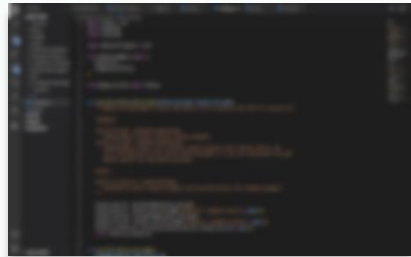
What products are each client **most likely to buy**?

2

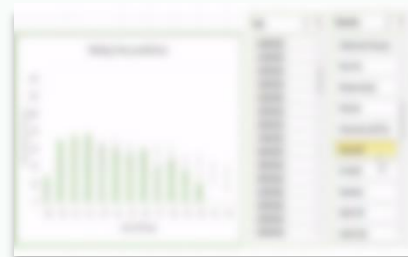
How to **leverage** this information to give strategic recommendations to the client?



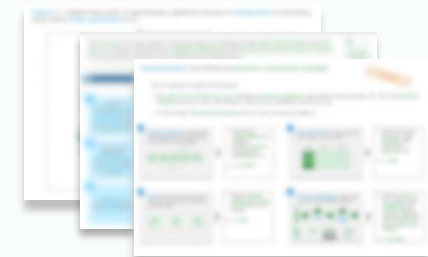
Deliverables



1. The **code** used to build the demo tool



2. A **visual dashboard** to present the results to the client



3. A “client-ready” deck of **PowerPoint slides**

Available data: you have access to **past transactions, clients, product with stock level** and information about **stores**



Goal: Recommend the best products of this company specializes in the sale of sports clothing to each of their clients for personalized marketing campaign



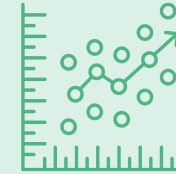
Find the best strategy to predict products of interest for each customer



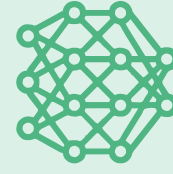
Data Analytics



Clustering



Regression



Deep Learning

...

You have access to **five datasets** to perform your analyses and build your models



transactions.csv



clients.csv



products.csv



stocks.csv



stores.csv



Relevant **external data** can complement those five datasets

Available data: past information are provided for each transactions in transactions.csv

1

ClientID	ProductID	StoreID	Quantity	SalesNetAmountEuro
6537585694072340000	892503538536458000	3306498059038630000	1	5.99
4683903573974630000	2280106402057210000	7822979941365010000	2	7.99
7339204980378490000	1246438640521450000	1218850731058950000	1	22.99
7771198051684050000	7364330090809850000	7716007625050230000	5	99.96
243477490322694000	535382046483972000	6262119512442080000	1	3.99
5373070388765250000	4574574996007880000	7822979941365010000	3	83.98
8841039056922080000	6535436876635010000	5766738857292150000	1	5.99
243477490322694000	8064013150495270000	6262119512442080000	1	2.99
1816503465925630000	880379940474282000	3306498059038630000	2	83.99
7918267621717290000	4274373756938080000	708932262589097000	1	14.99
3352816460031790000	6840454744607600000	1914079338419200000	3	39.98
212235332606805000	4216012665700460000	7123235743384990000	1	10.99
7325456381087170000	2001495150728660000	4320774246037820000	4	74.97
784139077380373000	4543350547232040000	5060113386821020000	2	7.99
4226302036211470000	17328091691254800	3306498059038630000	2	24.99
3783127575265090000	1993798909216080000	8556013948285210000	1	83.99
8114665738426690000	6473327096629740000	7200865266492670000	1	46.99

Available data: clients.csv contains details about the individual clients

2

ClientID	ClientSegment	ClientCountry	ClientOptINEmail	ClientOptINPhone	ClientGender	Age
4508698145640550000	LOYAL	USA	1	1	M	
2022746661324930000	INACTIVE_1Y	USA	0	1	F	
5794452591674300000	LOYAL	USA	1	1	F	
678556389231830000	LOYAL	USA	1	1	M	
877301557964624000	LOYAL	USA	1	1	F	
6676988281830110000	PROSPECT	USA	1	1	M	
1636102712153810000	LOYAL	USA	1	1	F	
1067727511891480000	LOYAL	USA	1	1	M	
8052621442524460000	LOYAL	USA	1	1	F	
1879338180562010000	INACTIVE_1Y	USA	1	1	F	
6393735473281830000	INACTIVE_1Y	USA	1	1	F	
8108569340379050000	INACTIVE_1Y	USA	0	1	M	
2843443116804930000	LOYAL	USA	1	1	M	30.0
8956762084833820000	INACTIVE_1Y	USA	1	1	F	
6568360953440530000	LOYAL	USA	0	1	F	
6164464346543320000	INACTIVE_1Y	USA	1	1	F	
7466342141455190000	LOYAL	USA	1	1	F	44.0

Available data: products.csv contains high level information about client's products

3

ProductID	Category	FamilyLevel1	FamilyLevel2	Universe
8588329712199130000	Basketball	Shoes	Adidas Harden Vol. 4	Women
2134543135517270000	Football	Shoes	Nike Mercurial Vapor	Women
7624925797167030000	Football	Ball	Nike Ordem V	Women
2270862675779330000	Football	Jersey	Adidas Home Jersey	Men
6504730903144740000	Football	Shorts	Adidas Squadra 21	Women
4829295361777980000	Football	Jersey	Adidas Home Jersey	Men
7935798579204670000	Football	Jersey	Adidas Home Jersey	Men
3698347634873450000	Football	Shoes	Nike Mercurial Vapor	Women
7713296683691040000	Football	Ball	Nike Ordem V	Women
43220326960179300	Football	Ball	Nike Ordem V	Women
2960383786256110000	Football	Shoes	New Balance Furon	Women
7767205741051530000	Football	Shoes	Adidas Predator	Men
5661242225760100000	Football	Ball	Adidas Telstar 18	Men
3850128663863540000	Football	Shoes	New Balance Furon	Women
3395043938592750000	Football	Shorts	Nike Dri-FIT	Women
7130074757728490000	Football	Shorts	Adidas Squadra 21	Men
2800704698504110000	Football	Ball	Puma Final 1	Women

Available data: stocks.csv provides stock by country

4

StoreCountry ▼	ProductID ▼	Quantity ▼
AUS	1284651161701380000	2.0
AUS	6076274819885030000	2.0
AUS	6019386668821120000	2.0
AUS	2122575437123250000	2.0
AUS	5901681811213090000	2.0
AUS	5195625195157270000	2.0
AUS	3850091287569970000	1.0
AUS	7723338481459020000	3.0
AUS	310504000978060000	2.0
AUS	1224535603186040000	1.0
AUS	3274981684168810000	1.0
AUS	7148728138092920000	3.0
AUS	3570508915410090000	5.0
AUS	9039906805668370000	2.0
AUS	4101272015341460000	3.0
AUS	8523563856621850000	1.0
AUS	7169400755031740000	4.0

Available data: stores.csv gives the country of each store

5

StoreID	StoreCountry
7793989562498220000	BRA
5301108587679200000	BRA
2908896877692950000	BRA
2718131790848670000	BRA
7819332417840560000	BRA
6524776804058170000	FRA
66413934723788800	FRA
4351033551716780000	FRA
8696981135006800000	FRA
7637657206478660000	FRA
3440206128053660000	FRA
4439061635768160000	FRA
5133673095652920000	FRA
395121715121348000	FRA
2353472391675350000	FRA
3189779657740820000	FRA
3645972967113440000	FRA

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6 Organization

eleven augmented proposal challenge: deliver a convincing commercial exchange to your prospect leveraging your dual expertise in business and data science



Exercise

Kick-off

- ✓ Find your group of 5 to 6 people

Project

- ✓ On your topic, put yourself in a data consultant's shoes during a 1st commercial exchange:
 - ⇒ What is the context of the company? (value chain, competition, technological trends, etc.)
 - ⇒ How could you generate added-value for them using potential of new technological enablers? (e.g., time-series, machine learning for tabular data, NLP, etc.)
 - ⇒ How would the project be structured? (roadmap, key steps, meetings with management, etc.)
 - ⇒ Can you make a first assessment of the feasibility of your solution? (data visualization, 1st demo, etc.)
 - ⇒ How would you leverage the results to provide key business insights?
- ✓ Synthesize your work in a commercial exchange (up to 20 slides)

Wrap-up

- ✓ Upload your work on the Sharepoint (presentation, code, etc: see details at the end of this presentation)
- ✓ Present to the jury for 15-20' minutes and prepare for questions (5'm)

Two subjects



The Endless Line

How to forecast waiting times in an amusement park?



The Next Purchase

How to recommend the best product?



Your mission: structure a data-augmented commercial exchange

Reminder



Context and objectives

- Key questions: What do you understand from the project? Why is the client's need challenging?
- Present your **understanding** of the situation
- Include the **process**, the **needs**, the potential **ROI**, etc.
- c. 2/3 slides



Key success factors

- what would be the reason of your success?
- Explain the reasons of success: strong **interaction** with client? Deep **expertise** in AI?
- c. 1 slide



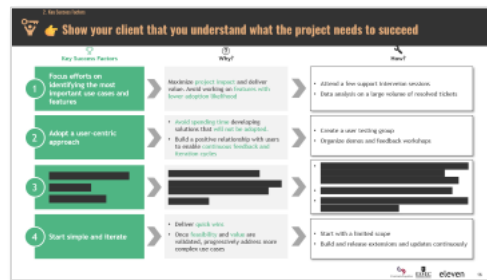
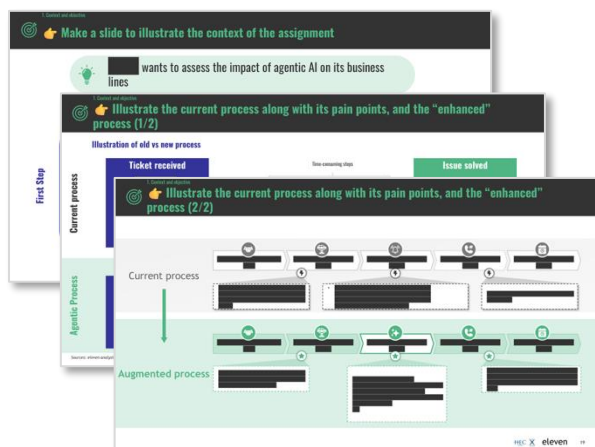
Methodology and planning

- what could you do for the potential client (implementation plan, ROI hypothesis, etc.)
- What you **could do**, **how**, what you have **already done**, any **relevant insight** to prove your expertise and **how much time** do you need **for the mission**
- Include a **demo**
- c. 6/8 slides



"Why you?"

- who are you and what have you done? Why are you relevant to solve the case?
- **Introduce yourself**, your **background** and explain **why you are relevant** for this mission
- c. 1 slide



Schedule for the five days: the timing is short, do not hesitate to split the work between the members of the team

	Monday	Tuesday	Wednesday	Thursday	Friday
AM	 Kick-off (9:00 am) In-depth understanding, and Q&A session (data + business)	Autonomous work	 Online Business Q&A • 1h timeslot per case	Autonomous work	 Final pitch • 20' per group • Planning on slack
PM	Autonomous work	Autonomous work	Autonomous work	Autonomous work	Closing session (2:00 pm)  Cocktail (~3:00)

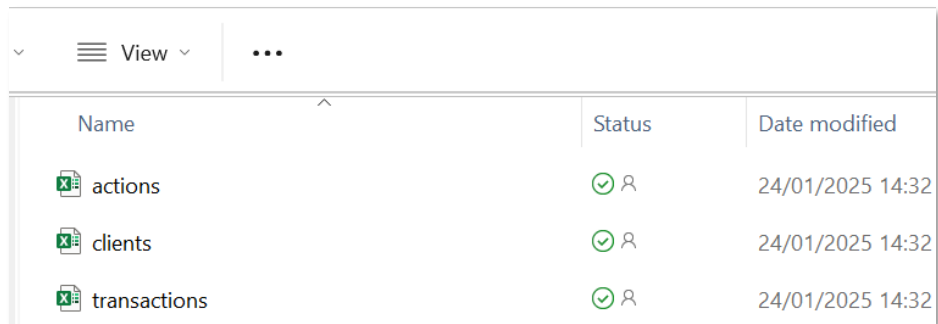
Your proposal will be assessed on your **business** and **technical** choices

Topic		Description	Points
Engagement		The engagement of the team during the exercise (how far you've gone, how autonomous you have been, how much you have asked questions when stuck, etc.)	2
Business aspect	Presentation quality	The quality of your final presentation : how professional it looks (slide quality), how clear and complete it is (storytelling), how pertinent your answers are, etc.	3
	Business methodology	The creativity and relevance of the methodology (i.e., scientific approach) you choose regarding the problem you try to solve, and the data provided, the business sense behind your methodology and the pragmatism of your presentation	6
Technical aspect	Technical choices	The explanation of your technical choices and your ability to present them in non-technical terms	3
	Model efficiency or analysis relevance	The performance of your model (specific to each case), the relevance for the problem in question , the quality of the analysis led	5
	Code good practices	Your code must be well structured , easy to run and easy to understand with clear readme and requirements.	1

Download instructions & submission process

How to download the datasets?

- Case #1: “The Endless Line” - [link here](#)
- Case #2: “The Next Purchase” - [link here](#)
- Password: dsba-2026-eleven-jP27e1G



Name	Status	Date modified
 actions	✓ 👤	24/01/2025 14:32
 clients	✓ 👤	24/01/2025 14:32
 transactions	✓ 👤	24/01/2025 14:32

Where to submit your work?

Each group will receive a link through Slack with a Sharepoint folder to submit their assignment (Presentation + code)



Download the datasets

Case #1: “The Endless Line”



[link here](#)

Case #2: “The Next Purchase”



[link here](#)

Password: dsba-2026-eleven-jP27e1G

Find your groups

1 - Endless Line	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6	Group 7
Student 1	Alexis Leclercq	Naël ARNOUX	Navneeth Rajesh	Ke Chen	Nicolas Perion	Charles de Cossé Briss	Javiera BAO
Student 2	Aris Fevre	Yurong WEN	Sarah Ela George	Haoju Li	Neil Dinia	Valentin Levy	Sanchay BHUTANI
Student 3	Camille Desurmont	Wei LUO	Joshua Alexander Ebenezer		Zihan Yang	Baptiste Noël	Sacha BOUBLIL
Student 4	Olga Ng	Faidon Kotsakis	Artur Garipov	Chien-Wei WENG	Matteo Couchoud	Lidia Pinto Guriel	Tommaso CAMPI
Student 5	Elene Wang	Yann Djouka	Alessandro Ivashkevich	Hangbo Yang	Marta Shkreli	Yingzhou Fang	Tamara ĐOKIC
Student 6	Enora Barbier		Mathilde Lafon	Lin Cai	Anas Khalil	Beatriz Passanezi	Nilay PURAYAR

2 - Your Next Purchase	Group 8	Group 9	Group 10	Group 11	Group 12	Group 13	Group 14
Student 1	Aashrith R	Modhura DAS	Xiangwen CHEN	Lolita Aboa	Zhe Wang	Ayushmaan Khalkho	Henri BOISSON
Student 2	Anirudh S	Alexia YE	Zihan LIN	Chamila Carémiaux		Sagar Vishishta	Alexandre LE SAUX
Student 3	Anupama Ajith	Jose MOLINA	Dan XIAO	Marc El Aakra	Quentin Rebut	Vishakh Shah	Gerrard O'BRIEN
Student 4	Satya Anirudh	Kris HUANG	Bingjing YUE	Lisa Nguyen	Haidar Bouaouiche	Andrew Bejjani	George EMMANUEL
Student 5	Jesslin Ann	Junyang ZHAO	Ying ZHOU	Léonie Willecomme	Sacha Koskas	Shan Cao	Nathan CLARK
Student 6	Siya Sinha	Faye PAGKALIWANGAN	Yang WAN	Wanchao Zhao	Yuhong Li	Monica Nathalie Bertoli	Tommaso CONTI

Join Us!



Interview process for **full-time positions**:

- ✓ **4-5** rounds
- ✓ Each round ~1 hour with **fit** and a data-oriented **case study**
- ✓ Apply now [here](#)



A question about the process? Reach out to one of DSB's alumni at eleven!



Baudoin
CS 25



Antoine Dargier
CS 23



Wallerand Peugeot
CS 22



Solène Leboucher
DSBA 25



Justine Cabon
ESSEC 23



Gaspard Defferrière
ESSEC 22